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JAGAPRIYAN B
CH.SC.U4CSE24216
LAB EXERCISE 11: PRINCIPAL COMPONENT ANALYSIS (PCA)
Topics Covered:
  Exercise 1: Real-Estate Analysis: Explaining Unsold Properties with PCA
  Exercise 2: Identifying Principal Component Dominant Variables from Factor Loadings Table
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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sn
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

print("="*70)
print("EXERCISE 11.1: MAGGIE'S REAL-ESTATE UNSOLD PROPERTY ANALYSIS")
print("="*70)
# Maggie wants insights into why properties managed by her company remain unsold > 6 months.
# Features: Price ($k), Age_of_Property (yrs), Distance_from_City (km),
# Maintenance_Cost ($/mo), Number_of_Rooms, Days_on_Market.
np.random.seed(42)
n_houses = 200

price = np.random.normal(450, 120, n_houses).clip(150, 900)
age = np.random.normal(18, 8, n_houses).clip(1, 45)
distance_city = np.random.normal(22, 10, n_houses).clip(2, 50)
maint_cost = 0.5 * price + np.random.normal(100, 30, n_houses)
rooms = np.random.choice([2, 3, 4, 5, 6], size=n_houses, p=[0.1, 0.3, 0.35, 0.15, 0.1])
# Target: Unsold for > 6 months (180 days)
days_on_market = (0.2 * price) + (1.5 * age) + (2.0 * distance_city) + np.random.normal(0, 20, n_houses)
unsold_over_6months = (days_on_market > 180).astype(int)

df_re = pd.DataFrame({
    'Price_k': price,
    'Age_Years': age,
    'Distance_City_km': distance_city,
    'Maint_Cost_Month': maint_cost,
    'Rooms': rooms,
    'Unsold_Over_6Mos': unsold_over_6months
})

print("Real Estate Dataset Sample:")
print(df_re.head())
print(f"\nUnsold (>6 months) distribution:\n{df_re['Unsold_Over_6Mos'].value_counts()}")

# Features to analyze
features = ['Price_k', 'Age_Years', 'Distance_City_km', 'Maint_Cost_Month', 'Rooms']
X = df_re[features]

# Step 1: Standardize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Step 2: Fit PCA
pca = PCA(n_components=len(features))
X_pca = pca.fit_transform(X_scaled)

print("\n--- Explained Variance by Principal Components ---")
explained_variance = pca.explained_variance_ratio_
cumulative_variance = np.cumsum(explained_variance)
for i, (ev, cv) in enumerate(zip(explained_variance, cumulative_variance)):
    print(f"PC{i+1}: {ev*100:.2f}% (Cumulative: {cv*100:.2f}%)")

# Principal Component Loadings (Weights)
df_loadings = pd.DataFrame(
    pca.components_[:3].T,
    index=features,
    columns=['PC1', 'PC2', 'PC3']
)
print("\n--- Component Loadings (Correlation of features with PCs) ---")
print(df_loadings.round(3))

# Insight Interpretation for Maggie:
print("\nInsights for Maggie:")
print("1. PC1 is heavily driven by Price and Maintenance Cost (Affordability Factor).")
print("2. PC2 is dominated by Age of Property and Distance to City Center (Location/Condition Factor).")
print("Properties far from city center and older properties with high maintenance remain unsold longest.")
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# Visualizing PC1 vs PC2
plt.figure(figsize=(8, 6))
scatter = plt.scatter(X_pca[:, 0], X_pca[:, 1], c=df_re['Unsold_Over_6Mos'], cmap='coolwarm', edgecolors='k', s=60)
plt.title("Real Estate PCA: PC1 vs PC2 (Unsold > 6 Months Highlighted)")
plt.xlabel(f"PC1 ({explained_variance[0]*100:.1f}% Variance Explained)")
plt.ylabel(f"PC2 ({explained_variance[1]*100:.1f}% Variance Explained)")
plt.colorbar(scatter, label='Unsold > 6 Months (1=Yes, 0=No)')
plt.grid(True)
plt.show()


print("\n" + "="*70)
print("EXERCISE 11.2: PRINCIPAL COMPONENT IDENTIFICATION FROM LOADINGS")
print("="*70)
# Exact Dataset Table given in Lab Manual (Page 52):
# Variable       | 1      | 2      | 3
# Climate        | 0.190  | 0.017  | 0.207
# Housing        | 0.544  | 0.020  | 0.204
# Health         | 0.782  | -0.605 | 0.144
# Crime          | 0.365  | 0.294  | 0.585
# Transportation | 0.585  | 0.085  | 0.234
# Education      | 0.394  | -0.273 | 0.027
# Arts           | 0.985  | 0.126  | -0.111
# Recreation     | 0.520  | 0.402  | 0.519
# Economy        | 0.142  | 0.150  | 0.239

manual_loadings = {
    'Variable': ['Climate', 'Housing', 'Health', 'Crime', 'Transportation',
                 'Education', 'Arts', 'Recreation', 'Economy'],
    'PC1': [0.190, 0.544, 0.782, 0.365, 0.585, 0.394, 0.985, 0.520, 0.142],
    'PC2': [0.017, 0.020, -0.605, 0.294, 0.085, -0.273, 0.126, 0.402, 0.150],
    'PC3': [0.207, 0.204, 0.144, 0.585, 0.234, 0.027, -0.111, 0.519, 0.239]
}

df_loadings_manual = pd.DataFrame(manual_loadings)
print("Principal Component Loadings Table from Manual (Page 52):")
print(df_loadings_manual)

# Identify which variable has maximum absolute loading for each Principal Component
print("\n--- Dominant Variable for each Principal Component ---")
for pc in ['PC1', 'PC2', 'PC3']:
    max_idx = df_loadings_manual[pc].abs().idxmax()
    dominant_var = df_loadings_manual.loc[max_idx, 'Variable']
    loading_val = df_loadings_manual.loc[max_idx, pc]
    print(f"{pc}: Most dominant variable is '{dominant_var}' with loading {loading_val}")

# Heatmap Visualization of Loadings
plt.figure(figsize=(7, 6))
loadings_matrix = df_loadings_manual.set_index('Variable')
sn.heatmap(loadings_matrix, annot=True, cmap='vlag', center=0, fmt='.3f')
plt.title("Principal Component Loadings Heatmap")
plt.show()
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